

Project: Student Dropout Prediction using Logistic Regression

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Student Dropout Prediction

Phase 1: Problem Understanding

Dataset Link: <https://www.kaggle.com/datasets/meharshanali/student-dropout-prediction-dataset>

1. Problem Statement

Student dropout is a major challenge for educational institutions, it affects student outcomes, institutional performance, and wastes resources spent on students who eventually leave mid-program.

Early identification allows institutions to intervene before a student actually withdraws through academic support, financial aid, or counseling.

Goal: Build a binary classification model (Logistic Regression) that predicts whether a student is at risk of dropping out, using their academic, demographic, socioeconomic, and enrollment-related information.

2. Target Variable

Variable: Dropout

Type: Binary(0/1)

Meaning: 0 = Did not drop out, 1 = Dropped out

Note: this target is imbalanced (~76% class 0, ~24% class 1). This will matter later when choosing evaluation metrics — accuracy alone will be misleading, so precision, recall, F1-score, and ROC-AUC will be used in Phase 6.

3. Input Features

Category	Features
Demographic	Age, Gender, Parental_Education
Socioeconomic	Family_Income, Internet_Acces, Part_Time_Job, Scholarship, Travel_Time_Minutes
Academic	GPA, Semester_GPA, CGPA, Attendance_Rate, Assignment_Delay_Days
Behavioral / Wellbeing	Study_Hours_per_Day, Stress_Index
Enrollment	Semester, Department
Identifier	student_ID it will not be used as a model feature

4. Real-World Application

A university could use this system as an early-warning tool to identify currently enrolled students who may be at higher risk of dropping out. Students flagged as high-risk could then be referred to academic advisors, counselors, or financial-aid staff for appropriate support. This would allow the institution to take proactive action instead of waiting until a student has already withdrawn.

Project Phase 2 : Dataset Collection & Exploration

2.1 Dataset Overview

The student dropout dataset contains academic, demographic, socioeconomic, behavioral, and enrollment-related information about students.

The purpose of this phase is to explore the dataset structure, understand the available features, identify the target variable, and examine the distribution of dropout cases before model development.

Import and Load Dataset

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
import pandas as pd

# Load the dataset
path = 'student_dropout_dataset_v3.csv'
df = pd.read_csv("/content/drive/MyDrive/student_dropout_dataset_v3.csv")

print("First 5 Rows:")
df.head(5)
```

First 5 Rows:

	Student_ID	Age	Gender	Family_Income	Internet_Access	Study_Hours_per_Day	Attendance_Rate	Assignment_Delay_Days	Travel_
0	1	22.1	Male	25000.0	Yes	3.36	86.1		2
1	2	20.7	Male	25000.0	Yes	4.30	68.0		2
2	3	22.4	Male	40183.0	Yes	4.40	70.9		0
3	4	24.4	Male	NaN	Yes	NaN	82.2		2
4	5	20.5	Female	25319.0	Yes	4.19	75.7		1

Next steps: [Generate code with df](#) [New interactive sheet](#)

Data Types

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 19 columns):
#   Column                                Non-Null Count  Dtype
---  ---                                -
0   Student_ID                            10000 non-null  int64
1   Age                                   10000 non-null  float64
2   Gender                                10000 non-null  object
3   Family_Income                         9500 non-null   float64
4   Internet_Access                       10000 non-null  object
5   Study_Hours_per_Day                   9500 non-null   float64
6   Attendance_Rate                       10000 non-null  float64
7   Assignment_Delay_Days                 10000 non-null  int64
8   Travel_Time_Minutes                   10000 non-null  float64
9   Part_Time_Job                         10000 non-null  object
10  Scholarship                           10000 non-null  object
11  Stress_Index                           9500 non-null   float64
12  GPA                                    10000 non-null  float64
13  Semester_GPA                          10000 non-null  float64
14  CGPA                                  10000 non-null  float64
15  Semester                               10000 non-null  object
16  Department                             10000 non-null  object
17  Parental_Education                    9489 non-null   object
18  Dropout                               10000 non-null  int64
dtypes: float64(9), int64(3), object(7)
memory usage: 1.4+ MB
```

Dataset Shape

```
print("Number of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])
```

Number of Rows: 10000
Number of Columns: 19

Dataset Columns

```
print("Dataset Columns:\n")

for i, column in enumerate(df.columns, start=1):
    print(f"{i}. {column}")
```

Dataset Columns:

```
1. Student_ID
2. Age
3. Gender
4. Family_Income
5. Internet_Access
6. Study_Hours_per_Day
7. Attendance_Rate
8. Assignment_Delay_Days
9. Travel_Time_Minutes
10. Part_Time_Job
11. Scholarship
12. Stress_Index
13. GPA
14. Semester_GPA
15. CGPA
16. Semester
17. Department
18. Parental_Education
19. Dropout
```

Numerical and Categorical Features

```
print("Numerical Features:")
print(df.select_dtypes(include='number').columns.tolist())

print("\nCategorical Features:")
print(df.select_dtypes(include='object').columns.tolist())
```

Numerical Features:
 ['Student_ID', 'Age', 'Family_Income', 'Study_Hours_per_Day', 'Attendance_Rate', 'Assignment_Delay_Days', 'Travel_Time_Minutes',
 Categorical Features:
 ['Gender', 'Internet_Access', 'Part_Time_Job', 'Scholarship', 'Semester', 'Department', 'Parental_Education']

Target Distribution

```
print("Dropout Distribution:")
print(df['Dropout'].value_counts())

print("\nDropout Percentage:")
print((df['Dropout'].value_counts(normalize=True) * 100).round(2))
```

Dropout Distribution:

```
Dropout
0    7646
1     2354
Name: count, dtype: int64
```

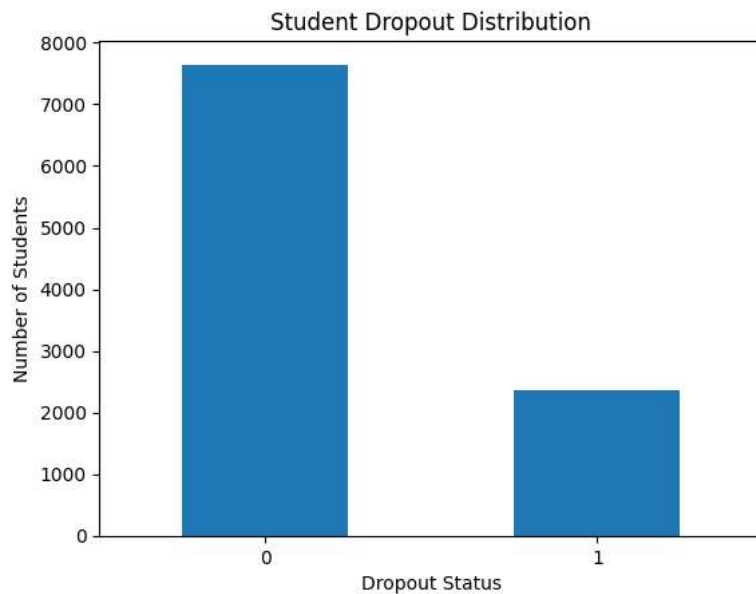
Dropout Percentage:

```
Dropout
0     76.46
1     23.54
Name: proportion, dtype: float64
```

```
import matplotlib.pyplot as plt

df['Dropout'].value_counts().plot(kind='bar')

plt.title('Student Dropout Distribution')
plt.xlabel('Dropout Status')
plt.ylabel('Number of Students')
plt.xticks(rotation=0)
plt.show()
```



✓ Project Phase 3 : Data Cleaning & Preprocessing

3.1 Data Cleaning and Preprocessing

Check Missing Values

```
missing_values = df.isnull().sum()

print("Missing values in each column:")
print(missing_values)
```

```
Missing values in each column:
Student_ID      0
Age             0
Gender          0
Family_Income   500
Internet_Access 0
Study_Hours_per_Day 500
Attendance_Rate 0
Assignment_Delay_Days 0
Travel_Time_Minutes 0
Part_Time_Job   0
Scholarship     0
Stress_Index    500
GPA             0
Semester_GPA    0
CGPA            0
Semester        0
Department      0
Parental_Education 511
Dropout         0
dtype: int64
```

Missing-Value Summary

```
missing_summary = pd.DataFrame({
    'Missing Values': df.isnull().sum(),
    'Missing Percentage': (df.isnull().mean() * 100).round(2)
})

missing_summary
```

	Missing Values	Missing Percentage	
Student_ID	0	0.00	
Age	0	0.00	
Gender	0	0.00	
Family_Income	500	5.00	
Internet_Access	0	0.00	
Study_Hours_per_Day	500	5.00	
Attendance_Rate	0	0.00	
Assignment_Delay_Days	0	0.00	
Travel_Time_Minutes	0	0.00	
Part_Time_Job	0	0.00	
Scholarship	0	0.00	
Stress_Index	500	5.00	
GPA	0	0.00	
Semester_GPA	0	0.00	
CGPA	0	0.00	
Semester	0	0.00	
Department	0	0.00	
Parental_Education	511	5.11	
Dropout	0	0.00	

Next steps: [Generate code with missing_summary](#) [New interactive sheet](#)

Handle the missing values

```
# Create a copy for preprocessing
df_clean = df.copy()

# Identify numerical and categorical columns
numeric_columns = df_clean.select_dtypes(include=['int64', 'float64']).columns
categorical_columns = df_clean.select_dtypes(include=['object']).columns

# Fill numerical missing values with the median
for column in numeric_columns:
    df_clean[column] = df_clean[column].fillna(df_clean[column].median())

# Fill categorical missing values with the mode
for column in categorical_columns:
    df_clean[column] = df_clean[column].fillna(df_clean[column].mode()[0])

print("Missing values after cleaning:")
print(df_clean.isnull().sum())
```

```
Missing values after cleaning:
Student_ID      0
Age             0
Gender          0
Family_Income   0
Internet_Access 0
Study_Hours_per_Day 0
Attendance_Rate 0
Assignment_Delay_Days 0
Travel_Time_Minutes 0
Part_Time_Job   0
Scholarship     0
Stress_Index    0
GPA             0
Semester_GPA    0
CGPA            0
Semester        0
Department      0
Parental_Education 0
```

```
Dropout          0
dtype: int64
```

Cleaning the Dataset Remove student_ID

```
# Remove identifier column
df_clean = df_clean.drop(columns=['Student_ID'])

print("Original shape:", df.shape)
print("Cleaned shape:", df_clean.shape)
```

```
Original shape: (10000, 19)
Cleaned shape: (10000, 18)
```

Check Duplicates

```
print("Number of duplicate rows:", df_clean.duplicated().sum())
```

```
Number of duplicate rows: 0
```

Encoding Categorical Variables

```
categorical_columns = df_clean.select_dtypes(include=['object']).columns.tolist()

print("Categorical columns:")
print(categorical_columns)
```

```
Categorical columns:
['Gender', 'Internet_Access', 'Part_Time_Job', 'Scholarship', 'Semester', 'Department', 'Parental_Education']
```

```
df_encoded = pd.get_dummies(
    df_clean,
    columns=categorical_columns,
    drop_first=True,
    dtype=int
)

print("Shape after encoding:", df_encoded.shape)
```

```
Shape after encoding: (10000, 25)
```

```
df_encoded.head()
```

	Age	Family_Income	Study_Hours_per_Day	Attendance_Rate	Assignment_Delay_Days	Travel_Time_Minutes	Stress_Index	GPA	Semester
0	22.1	25000.0	3.36	86.1	2	20.4	5.5	0.96	
1	20.7	25000.0	4.30	68.0	2	44.0	6.8	1.28	
2	22.4	40183.0	4.40	70.9	0	48.9	5.5	1.68	
3	24.4	29740.5	4.00	82.2	2	38.6	5.5	1.78	
4	20.5	25319.0	4.19	75.7	1	23.0	7.0	1.48	

5 rows × 25 columns

Separate Features and Target

```
X = df_encoded.drop(columns=['Dropout'])
y = df_encoded['Dropout']

print("Features (X) shape:", X.shape)
print("Target (y) shape:", y.shape)
```

```
Features (X) shape: (10000, 24)
Target (y) shape: (10000,)
```

Final Dataset

```
print("Final feature dataset:")
print("X shape:", X.shape)
print("y shape:", y.shape)

print("\nFinal feature columns:")
print(X.columns.tolist())
```

Final feature dataset:

X shape: (10000, 24)

y shape: (10000,)

Final feature columns:

['Age', 'Family_Income', 'Study_Hours_per_Day', 'Attendance_Rate', 'Assignment_Delay_Days', 'Travel_Time_Minutes', 'Stress_Index', 'GPA', 'Dropout']

Phase 4 : Exploratory Data Analysis

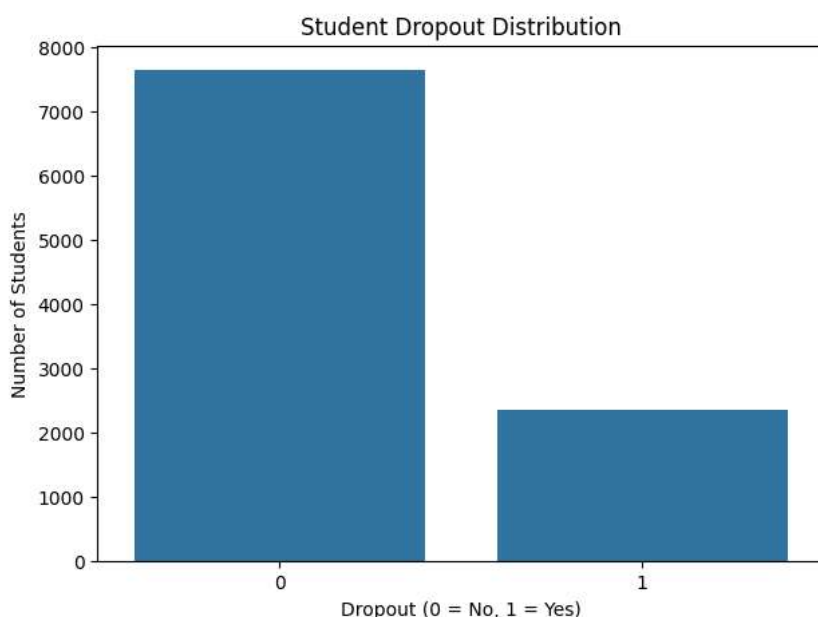
Dropout Distribution

```
import matplotlib.pyplot as plt
import seaborn as sns

# Dropout distribution
dropout_counts = df_clean['Dropout'].value_counts().sort_index()

plt.figure(figsize=(7, 5))
sns.barplot(x=dropout_counts.index, y=dropout_counts.values)

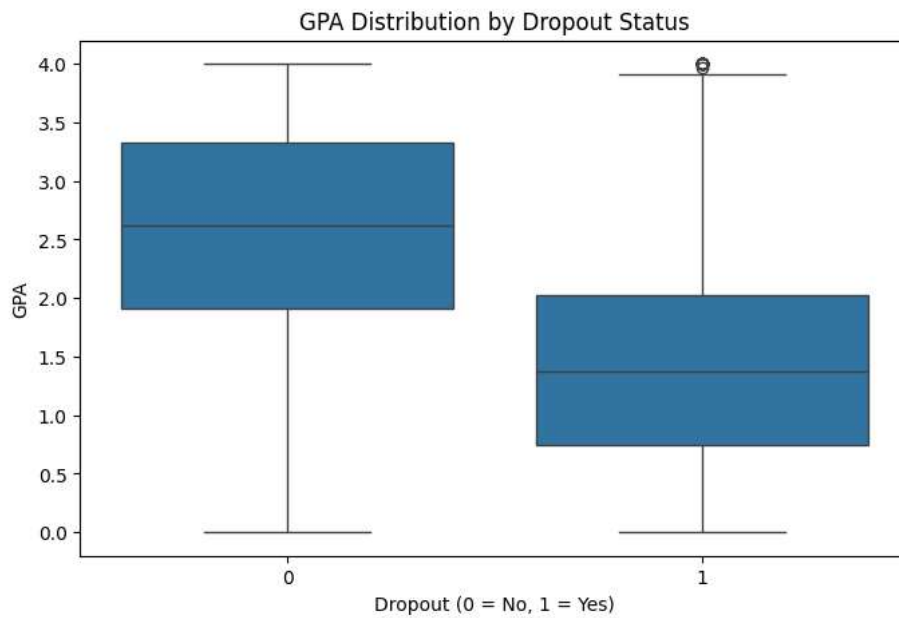
plt.title('Student Dropout Distribution')
plt.xlabel('Dropout (0 = No, 1 = Yes)')
plt.ylabel('Number of Students')
plt.show()
```



Academic Performance vs Dropout

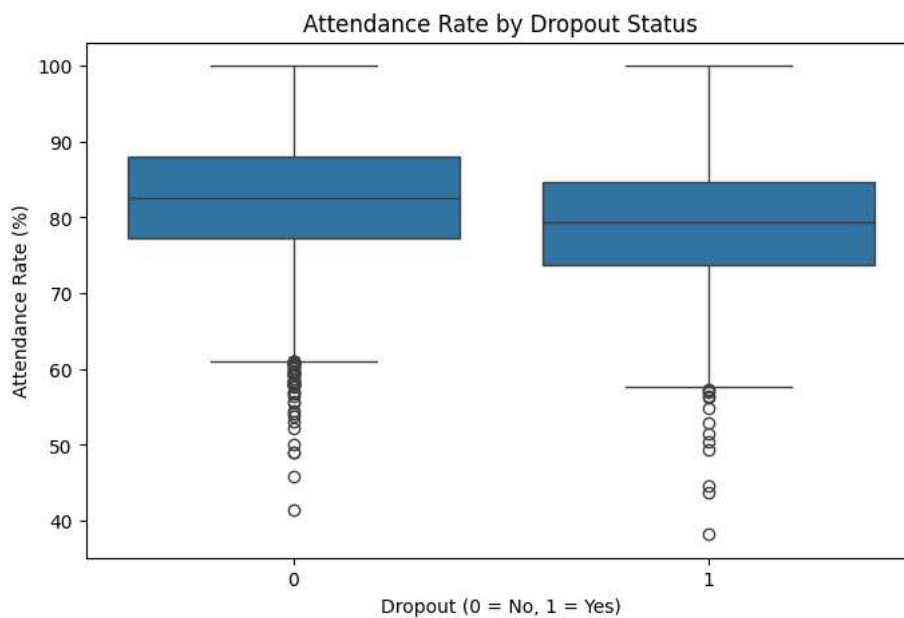
```
#GPA by dropout status
plt.figure(figsize=(8, 5))
sns.boxplot(x='Dropout', y='GPA', data=df_clean)

plt.title('GPA Distribution by Dropout Status')
plt.xlabel('Dropout (0 = No, 1 = Yes)')
plt.ylabel('GPA')
plt.show()
```



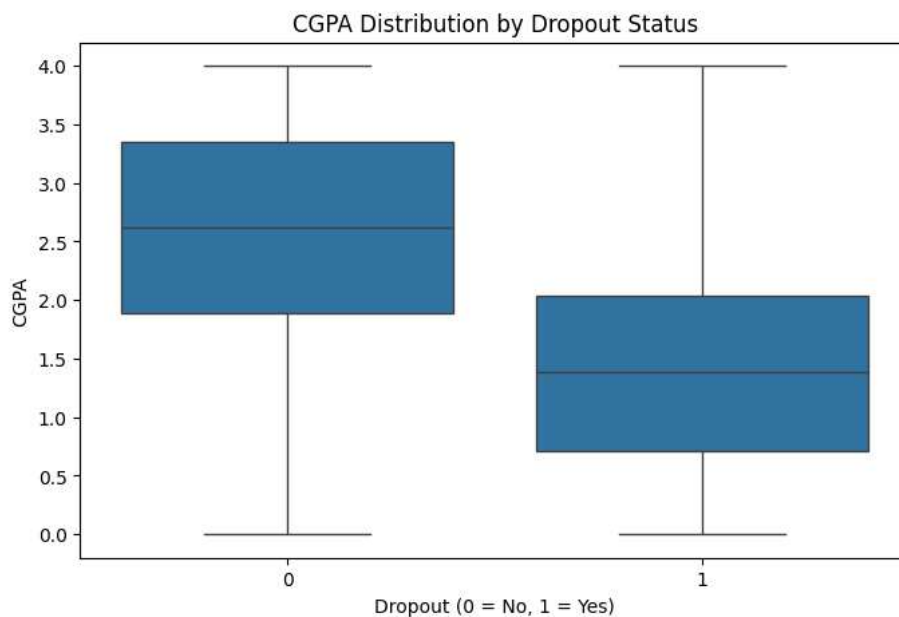
```
#Attendance vs Dropout
plt.figure(figsize=(8, 5))
sns.boxplot(x='Dropout', y='Attendance_Rate', data=df_clean)

plt.title('Attendance Rate by Dropout Status')
plt.xlabel('Dropout (0 = No, 1 = Yes)')
plt.ylabel('Attendance Rate (%)')
plt.show()
```



```
#CGPA vs Dropout
plt.figure(figsize=(8, 5))
sns.boxplot(x='Dropout', y='CGPA', data=df_clean)

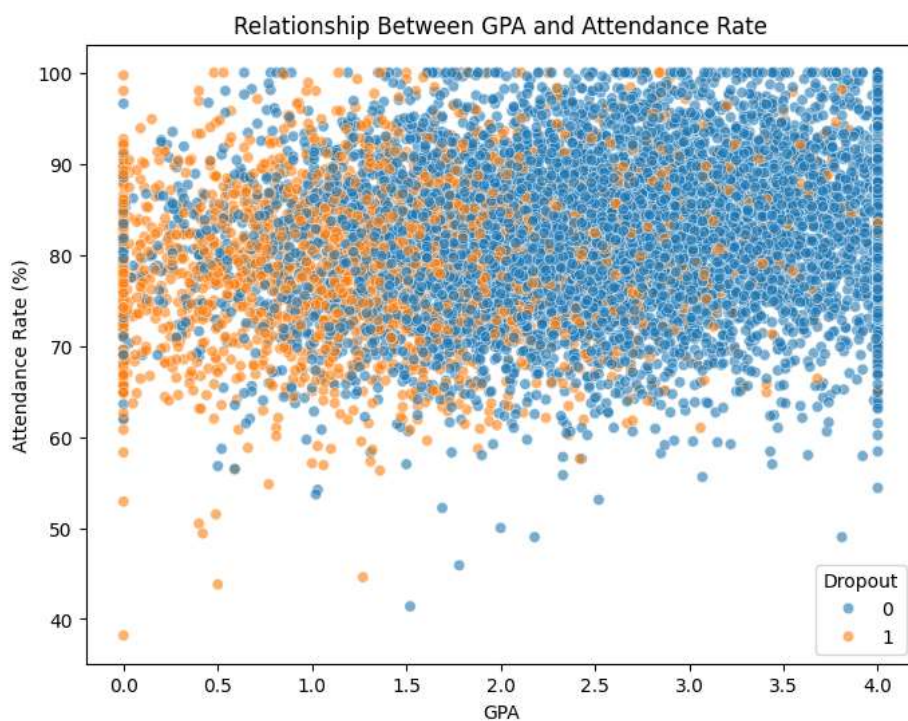
plt.title('CGPA Distribution by Dropout Status')
plt.xlabel('Dropout (0 = No, 1 = Yes)')
plt.ylabel('CGPA')
plt.show()
```



Feature Relationships

```
#GPA vs Attendance
plt.figure(figsize=(8, 6))
sns.scatterplot(
    data=df_clean,
    x='GPA',
    y='Attendance_Rate',
    hue='Dropout',
    alpha=0.6
)

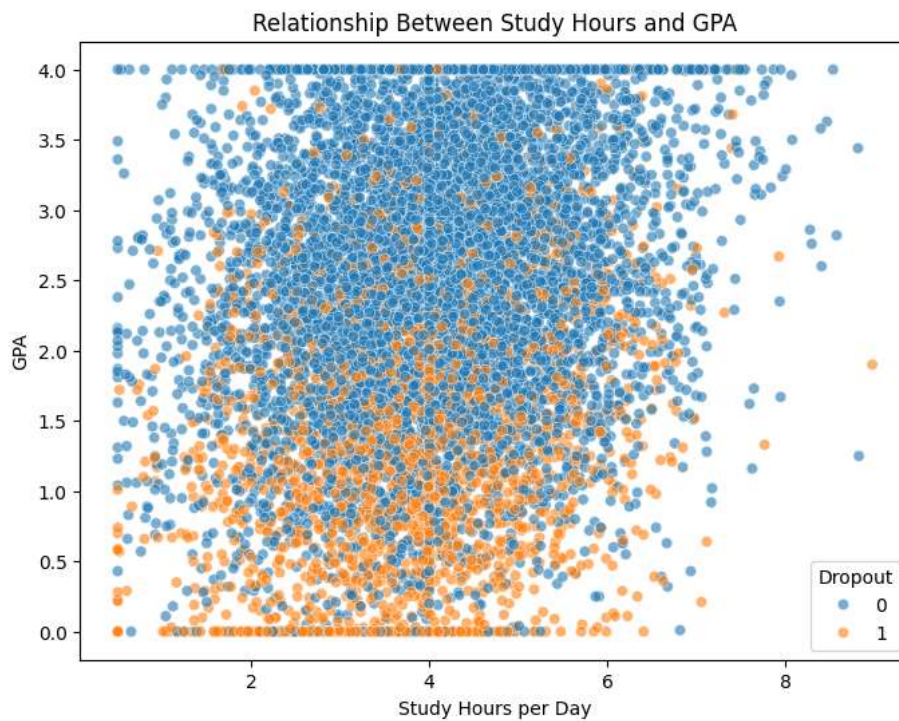
plt.title('Relationship Between GPA and Attendance Rate')
plt.xlabel('GPA')
plt.ylabel('Attendance Rate (%)')
plt.show()
```



```
#Study Hours vs GPA
plt.figure(figsize=(8, 6))
```

```
sns.scatterplot(
    data=df_clean,
    x='Study_Hours_per_Day',
    y='GPA',
    hue='Dropout',
    alpha=0.6
)

plt.title('Relationship Between Study Hours and GPA')
plt.xlabel('Study Hours per Day')
plt.ylabel('GPA')
plt.show()
```



Numerical Feature Correlation

```
# Correlation Heatmap
# Select numerical features
numeric_df = df_clean.select_dtypes(include=['int64', 'float64'])

plt.figure(figsize=(12, 8))
sns.heatmap(
    numeric_df.corr(),
    annot=True,
    fmt='.2f',
    cmap='coolwarm',
    center=0
)

plt.title('Correlation Between Numerical Features')
plt.show()
```



```
comparison = df_clean.groupby('Dropout')[
    ['GPA',
     'Semester_GPA',
     'CGPA',
     'Attendance_Rate',
     'Study_Hours_per_Day',
     'Stress_Index']
].mean()

comparison.round(2)
```

	GPA	Semester_GPA	CGPA	Attendance_Rate	Study_Hours_per_Day	Stress_Index
Dropout						
0	2.58	2.57	2.56	82.48	4.07	5.27
1	1.43	1.44	1.44	79.31	3.82	6.28

Important Observations

1. The dataset contains both dropout and non-dropout students, with

Based on the exploratory analysis:

students in the non-dropout group.

2. Dropout students generally show lower academic performance than students, particularly in GPA and Semester GPA.

3. Dropout students have a lower average number of study hours per with non-dropout students.

4. The average Stress Index is higher among dropout students, sugg

1. The dataset contains both dropout and non-dropout students, with more students in the non-dropout group.

2. Dropout students generally show lower academic performance than non-dropout students, particularly in GPA and Semester GPA.

3. Dropout students have a lower average number of study hours